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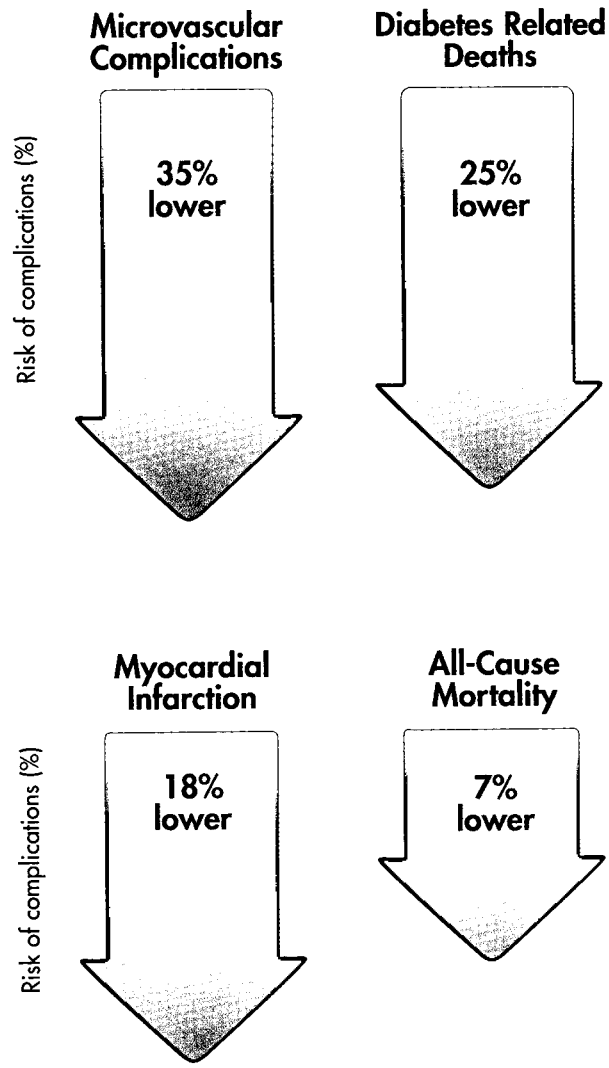
Decrease in Risk of Complications [12]

A 1% drop in A1C level can reduce the incidence of long-term complications from diabetes.[12]



Data from the ADA position statement on the United Kingdom Prospective Diabetes Study.[12]

Complications Due to Diabetes

Long-Term Health Problems

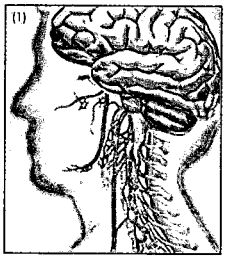
Continued elevated glucose levels cause damage to small blood vessels as well as large arteries. Macrovascular complications include coronary heart disease, peripheral vascular disease, and stroke.[9] Microvascular complications (retinopathy, nephropathy and possibly neuropathy) decreased by 35% with a 1% reduction in A1C in the UKPDS study.[12]

UKPDS has shown that by the time diabetes is diagnosed, approximately half of the people with type 2 diabetes already have evidence of tissue damage.[13] Therefore, it is necessary to initiate care to obtain near-normal blood glucose and blood pressure levels in an effort to reduce the risk of diabetic complications.[14]

Nephropathy

Diabetic nephropathy occurs in 20-40% of patients with diabetes and is the single leading cause of end-stage renal disease[2]. It is a microvascular complication where the glomeruli's (filtering units of the kidney) function to filter blood becomes compromised. Microalbuminuria (elevated levels of albumin in the urine) has been shown to be the earliest indicator of diabetic nephropathy.

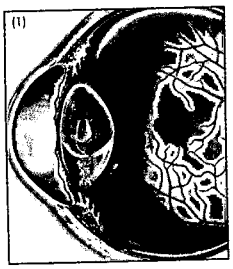
Neuropathy



Classified as a microvascular complication,[8] diabetic neuropathies are divided into diffuse or focal neuropathies.[2] Diabetic neuropathies are heterogeneous with diverse clinical manifestations. Most common among the neuropathies are chronic sensorimotor distal symmetric polyneuropathy (DPN) and autonomic neuropathy. Up to 50% of DPN may be asymptomatic and patients are at risk of insensate injury to their feet.[2] The symptoms and signs of autonomic dysfunction should be explored carefully during the physical exam. Key clinical manifestations of autonomic neuropathy include resting tachycardia, exercise intolerance, orthostatic hypotension, constipation, gastroparesis, erectile dysfunction, sudomotor dysfunction, impaired neurovascular function, "brittle diabetes," and hypoglycemic autonomic failure.[2]

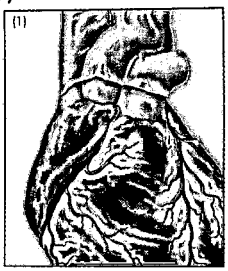
Images Adapted from:
- Netter FH.[1]
- Runge MS. et al.[15]
- Netter FH.[16]

Retinopathy



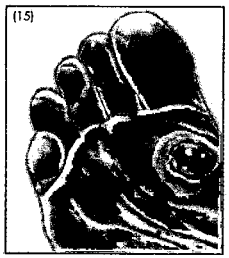
Diabetic retinopthy, which is found in both type 1 and type 2 diabetes, is the most frequent cause of new cases of blindness among adults aged 20-74 years.[2] Nourishment of blood vessels of the retina is severely compromised, triggering growth of new, extremely fragile blood vessels. Growth, swelling and leakage from these new vessels characterize this microvascular complication. Retinopathy is estimated to take at least 5 years to develop after onset of hyperglycemia.[2]

Cardiovascular Disease



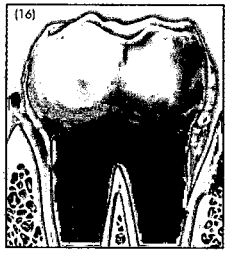
Cardiovascular disease includes coronary artery disease, peripheral vascular disease, and stroke.[9] Accounting for 65% of deaths in patients with diabetes, the risk of death from cardiovascular disease is twice as high for males with diabetes than the non-diabetic male, and up to four times higher among females with diabetes than the non-diabetic female.[8] Cardiovascular disease is the major cause of morbidity and mortality for individuals with diabetes and is the largest contributor to the direct and indirect costs of diabetes.[2]

Foot Care



Foot ulceration and amputation are common complications in diabetes. 15% of patients diagnosed with diabetes will develop foot ulcers with 20% of that population undergoing amputation. Peripheral neuropathy, altered biomechanics and peripheral arterial disease are all risk conditions associated with foot-related diabetic complications. All individuals with diabetes should receive an annual foot examination. Primary focal points to assess include protective sensation, foot structure and biomechanics, vascular status and skin integrity. Patients with one or more high risk factors should be evaluated more frequently and those with neuropathy should undergo a visual foot inspection at each visit with their health care professional.[2]

Periodontal Disease



Periodontal and dental disease are prevalent in people with diabetes. A number of conditions can occur including abscesses, gingivitis, and severe periodontitis with tooth loss. Commonly seen oral conditions with poorly controlled diabetes include candidiasis, lichen planus, xerostomia, burning mouth syndrome, angular cheilitis and taste dysfunction. The importance of proper oral hygiene should be reinforced.[8]

Maintaining Control of Diabetes [2]

People with diabetes can help prevent or delay problems by keeping blood glucose, blood pressure, and cholesterol under control and getting regular medical care.

People with diabetes need to work with a health care team to maintain a balance between medication, blood glucose monitoring, diet, and exercise.

The following should be done everyday to help control diabetes:

- Follow a healthy meal plan
- Become more physically active
- Maintain proper oral hygiene
- Refrain from smoking
- Monitor feet for cuts, blisters, or sores
- Take diabetes medication as needed
- Monitor blood glucose

Recommended Blood Pressure and Lipid Guidelines [2]

Assessment	Goal	Measure
Blood Pressure Goal (BP)	<130/80 mm Hg	Measure at every routine visit
Low-Density Lipoprotein (LDL)	<100 mg/dl	Most adults – measure annually Low risk – measure every 2 years
High-Density Lipoprotein (HDL) **MALES**	>40 mg/dl	Most adults – measure annually Low risk – measure every 2 years
High-Density Lipoprotein (HDL) **FEMALES**	>50 mg/dl	Most adults – measure annually Low risk – measure every 2 years
Triglycerides (TG)	<150 mg/dl	Most adults – measure annually Low risk – measure every 2 years

Diabetes Management Treatment Goals

Diagnostic Criteria for Diabetes* [2,7]

Category	Fasting Plasma Glucose	Casual Plasma Glucose	75g Oral Glucose Tolerance Test
Non-critically ill	FPG<100 mg/dl	–	2-hr PG <140 mg/dl
IFG/IGT ("Pre-Diabetes")	100 - 125 mg/dl	–	2-hr PG 140 - 199 mg/dl
Diabetes	FPG ≥126 mg/dl	PG ≥200 mg/dl plus symptoms	2-hr PG ≥200 mg/dl

* Must be confirmed

ADA Standards of Care* [2]

Examination	Frequency/Recommendation
A1C	2-4 Times a year
Dilated and Comprehensive Eye Exam	Yearly
Lipid profile	At least yearly & more often if needed to achieve goal
Blood pressure	Each visit
Microalbuminuria	Yearly
Serum creatinine	At least yearly
Asprin prophylaxis	If indicated
Influenza vaccine	Yearly
DPN (distal symmetric polyneuropathy)	Yearly
Foot	Yearly
Preconception care	Yes
Smoking cessation	Yes

* Individual patient frequency/recommendation may differ

Symptoms and Risk Factors of Diabetes

Type 1 Diabetes

Type 1 Diabetes is characterized by an absolute deficiency in insulin secretion, due to a cellular-mediated autoimmune destruction of the pancreatic ß-cells.[7] This destruction is likely mediated by auto-antibodies to these cells through a process referred to as autoimmunity. With the rapid onset of type 1 diabetes, some signs and symptoms may appear abruptly and rapidly increase in severity.[8]

Type 2 Diabetes

Type 2 Diabetes is a progressive, complex metabolic disorder. It is characterized by coexisting defects of multiple organ sites including insulin resistance in muscle and adipose tissue, a progressive decline in pancreatic insulin secretion, unrestrained hepatic glucose production, and other hormonal deficiencies. Patients often develop type 2 diabetes 9 to 12 years before the disease is diagnosed.[7] Because type 2 diabetes usually develops more slowly, it may lead to insidious development of barely noticeable symptoms.[8]

Symptoms of Diabetes [7,8]

- Polyuria
- Polyphagia
- Polydypsia
- Weight loss
- Blurred vision
- Fatigue
- Itching of the skin
- Wounds that are slow to heal

Risk Factors for Pre-diabetes and Diabetes Mellitus [7]

- Family history of diabetes
- Cardiovascular disease
- Overweight or obesity
- Sedentary lifestyle
- Latino/Hispanic, Non-Hispanic black, Asian American, Native American or Pacific Islander ethnicity
- Previously identified impaired glucose tolerance or impaired fasting glucose
- Hypertension
- Increased levels of triglycerides, low concentrations of high-density lipoprotein cholesterol, or both
- History of gestational diabetes
- History of delivery of an infant with a birth weight over 9 pounds
- Polycystic ovary syndrome
- Psychiatric illness

Common Terms and Definitions

- **Pre-diabetes**
  A condition in which blood glucose levels are higher than normal but not high enough for a diagnosis of diabetes.[9]
- **Blood glucose**
  The level of glucose found in the bloodstream. It is measured in milligrams in a deciliter, or mg/dl.[9,10]
- **FPG (fasting plasma glucose)**
  The blood glucose level after abstaining from eating for 8 to 12 hours (usually overnight).[9]
- **PPG (postprandial plasma glucose)**
  The blood glucose level taken 1 to 2 hours after a meal.[9]
- **IFG (impaired fasting glucose)**
  A condition in which a blood glucose test, taken after an 8 to 12 hour fast, shows a level of glucose higher than normal but not high enough for a diagnosis of diabetes. IFG, also called pre-diabetes, is an FPG level of 100 mg/dl to 125 mg/dl.[9]
- **IGT (impaired glucose tolerance)**
  A condition in which blood glucose levels are higher than normal but are not high enough for a diagnosis of diabetes. IGT, also called pre-diabetes, is a level of 140 mg/dl to 199 mg/dl 2 hours after the start of an oral glucose tolerance test.[9]
- **Hemoglobin A1C**
  Hemoglobin A1C reflects the mean blood glucose concentration over the preceding 2 to 3 months.[9]
- **Insulin**
  A hormone secreted by the beta cells of the pancreas that controls the metabolism and cellular uptake of sugars, proteins, and fats.[10]
- **Insulin analog**
  Insulin analogs are produced by recombinant DNA technology, in which certain amino acids in the insulin molecule have been modified.[9,11]
- **Basal rate**
  A steady rate of low levels of longer-acting insulin.[9]
- **Bolus**
  An extra amount of insulin taken to cover an expected rise in blood glucose, often related to a meal or snack.[9]

Targets for Glycemic Control - Inpatient [2,7]

	American Diabetes Association (ADA)	American Association of Clinical Endocrinologists (AACE)
Critically ill	110 mg/dl and <140 mg/dl	80 - 110 mg/dl
Non-critically ill	Fasting: <126 mg/dl Random: <180 - 200 mg/dl	Preprandial: <110 mg/dl Peak PPG: <180 mg/dl

Targets for Glycemic Control - Outpatient [2,7]

Assessment	ADA Goal	AACE Goal
Hemoglobin A1C	<7.0%*	≤6.5%
Fasting/prepandial Blood sugars	70 - 130 mg/dl	<110 mg/dl
2 hour PPG	<180 mg/dl	<140 mg/dl

* For non-pregnant adults

A1C Level and Mean Plasma Glucose Level Correlations [2]

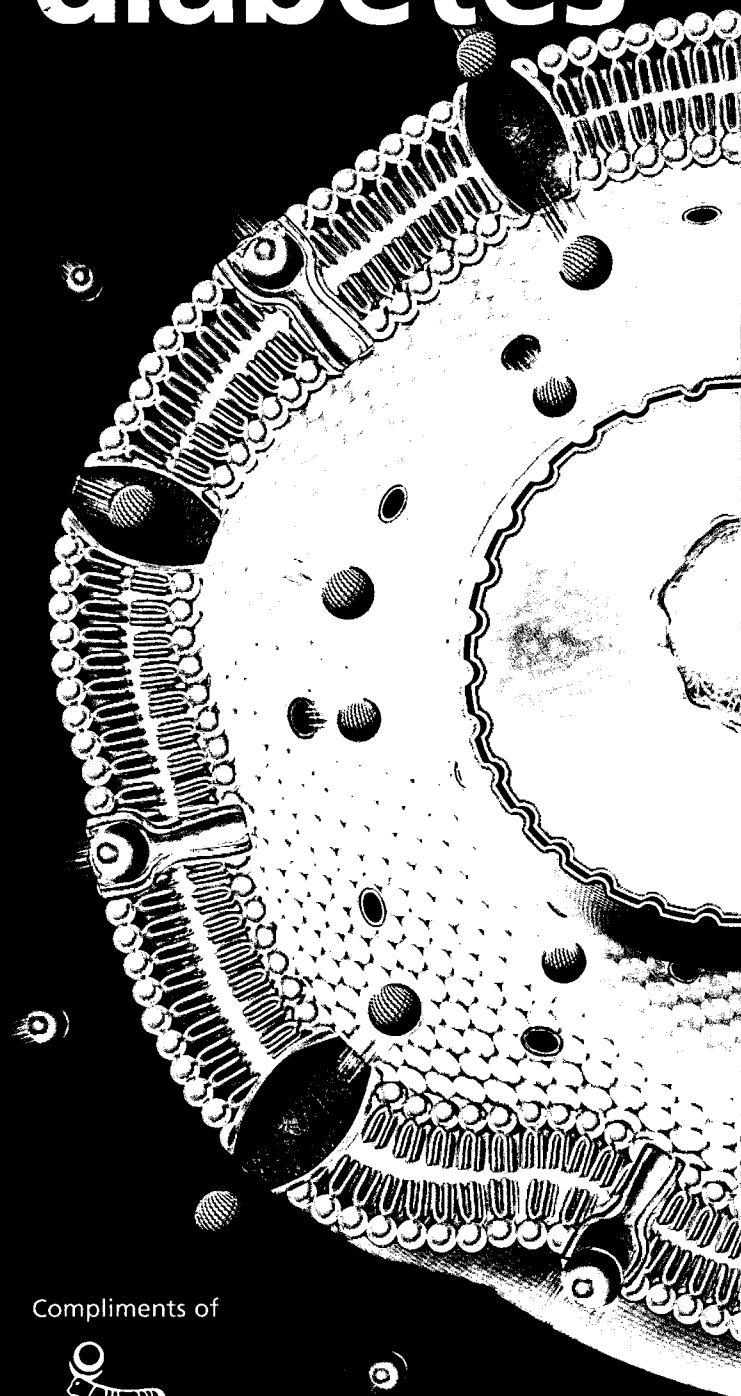
Hemoglobin A1C	Mean Plasma Glucose Value
6%	135 mg/dl
7%	170 mg/dl
8%	205 mg/dl
9%	240 mg/dl
10%	275 mg/dl
11%	310 mg/dl
12%	345 mg/dl

ADA Screening for Diabetic Nephropathy [2,7]

Test	Normal	Microalbumin	Macroalbumin
Random urine	<30 µg/mg creatinine	30 - 299 µg/mg creatinine	≥300 µg/mg creatinine

understanding

diabetes



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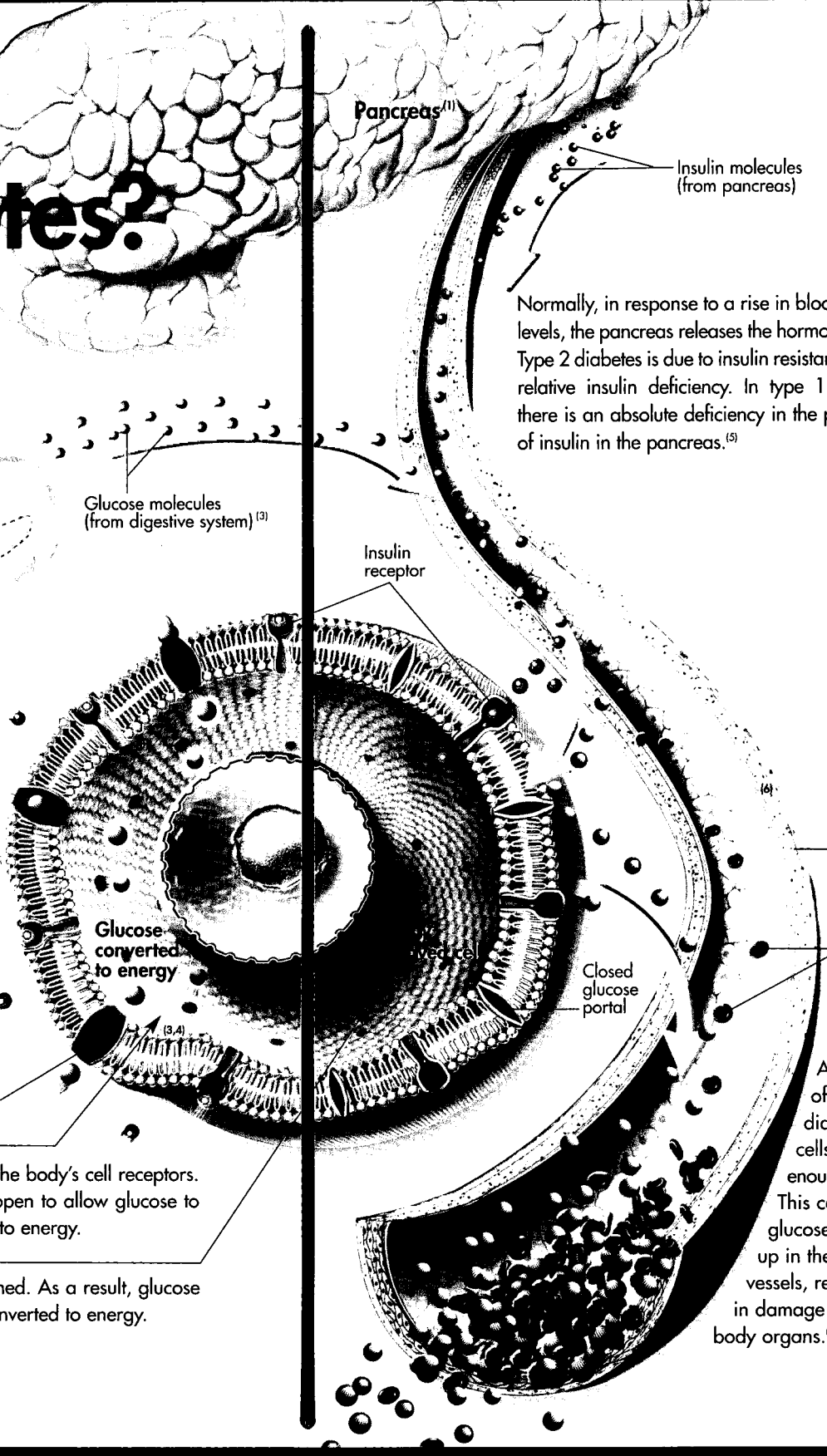
diabetes?

Diabetes is a chronic, metabolic disorder characterized by resistance to the action of insulin, insufficient insulin secretion, or both. The American Diabetes Association has established four clinical classes for use in diagnosing and determining the proper disease treatment plan.

Type 1 diabetes, which results from ß-cell destruction, usually leads to absolute insulin deficiency.[2]

Type 2 diabetes, the most common classification of diabetes, results from a progressive insulin secretory defect on the background of insulin resistance.[2]

Gestational diabetes mellitus encompasses any degree of glucose intolerance with onset or first recognition during pregnancy.[2] Finally, the fourth classification includes all **Other specific types of diabetes** due to other causes, including genetic defects in insulin action, diseases of the exocrine pancreas, and drug or chemical-induced.[2] Hyperglycemia occurs in all patients with diabetes.[2]



Normally, in response to a rise in blood glucose levels, the pancreas releases the hormone insulin. Type 2 diabetes is due to insulin resistance and or relative insulin deficiency. In type 1 diabetes, there is an absolute deficiency in the production of insulin in the pancreas.[5]

Normal Body Cell
Normally, insulin molecules bind to the body's cell receptors. When activated by insulin, portals open to allow glucose to enter the cells, where it is converted to energy.

Body Cell with Diabetes
Without insulin, portals are not opened. As a result, glucose is not able to enter the cells to be converted to energy.

As a result of type 2 diabetes, cells don't get enough energy. This causes glucose to build up in the blood vessels, resulting in damage to all body organs.[3]

Images Adapted from:
- Netter FH.[1]
- Hole JW Jr.[3]
- LeRoith D. et al.[4]
- Hammersen F.[6]